1. Which sorting algorithm performed better? Explain your observation.
   The Quick Sort algorithm is better than the bubble sort algorithm because the time complexity for sorting using Quick Sort is O(nlog(n)) while it is O(n^2) for bubble sort.
2. Which function consumed the maximum execution time?
   The partition function conserved the maximum execution time.
3. How does execution time change with increasing input size?
   For quicksort the execution time increased linearly upon increase of input size while it increased in order of n^2 for Bubble sort
4. How did compiler optimizations (-O2 and -O3) improve performance?
   The compiler optimizations O2 and O3 improve performance by reducing the function calls and eliminating redundant operations and generating more efficient machine code
5. Which algorithm would you recommend for large datasets? Justify your answer.
   For large datasets I would recommend Quick sort because Quick sort is faster than bubble sort and it does not increase exponentially upon an increase in data.

For 100000 random Values

	Quick Sort	Bubble Sort
Execution Time	0.008436173	16.268206703
CPU Cycles	0.008436173	71447462025
Instructions	26390915	52522650964
IPC	1.1	0.7
Cache References	126496	378278402
Cache Misses	7193	1999428
Hotspot Function	partition	partition